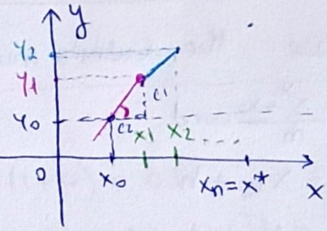
Introduction to numerical methods

DEplot → graph of a solution / an orbit → gives approximation.

$$\boxed{y' = g(x, y)}$$



Def. A solution is a function

$\varphi : I \to \mathbb{R}$ s.t. I = interval, $\varphi \in C^1$

and $\varphi'(x) = g(x, \varphi(x))$, $(\forall)\, x \in I$

We consider the IVP

(1): $y' = g(x, \varphi(x))$, $y(x_0) = y_0$, where x_0, y_0 are fixed.

Theorem 1: Assume $g \in C^1$ function. Let $(x_0, y_0) \in \mathbb{R}^2$. Then the IVP (1) has a unique solution, denoted with:

$$\varphi : I \to \mathbb{R}, \quad x_0 \in I$$

Given: The IVP (1): g, x_0, y_0

$x^* > 0$ s.t. $[x_0, x^*] \subset I$.

a partition of the interval $[x_0, x^*]$

$$x_0 < x_1 < x_2 < \dots < x_m = x^*$$

m = the nr. of steps needed to cover the interval

The aim: To find $y_1, y_2, \dots, y_m \in \mathbb{R}$ s.t.

$$y_1 \approx \varphi(x_1)$$
$$y_2 \approx \varphi(x_2)$$
$$\vdots$$
$$y_m \approx \varphi(x_m)$$

↓ "good approx."

Interpret: $(x_0, y_0) \in G_\varphi$

(x_k, y_k) are sufficiently close to G_φ, $k = \overline{1,m}$.

The Euler numerical formula to find the values $y_1, \dots, y_m$.

→ step 1: Let $m_0 = g(x_0, y_0)$ be the slope of the direction field in (x_0, y_0)

$$m_0 = \frac{c_1}{c_2} = \frac{y_1 - y_0}{x_1 - x_0} \Rightarrow y_1 = y_0 + m_0(x_1 - x_0)$$

$$\Rightarrow y_1 = y_0 + (x_1 - x_0) \cdot \underbrace{g(x_0, y_0)}_{m_0}$$

→ Step 2: $u_1 = g(x_1, y_1)$

$y_2 = y_1 + (x_2 - x_1) \cdot g(x_1, y_1)$

...

→ Step k+1: $\boxed{y_{k+1} = y_k + (x_{k+1} - x_k) \cdot g(x_k, y_k)}$

Particular case: the Euler's numerical formula with constant step size.

FOR EXAM

Let $h = \frac{x^* - x_0}{m}$ and

$\begin{cases} x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}$ $k = \overline{0, m-1}$, where x_0, y_0 are given.

Ex: We consider the IVP $\begin{cases} y' = 1 + xy^2 \\ y(0) = 0 \end{cases}$ whose unique solution is denoted by φ. Write the Euler's numerical formula with constant step size $h > 0$ for this IVP.

Now take $h = 0.1$. Find the nr. of steps to reach $x^* = 1$. Compute approx. values for $\varphi(0.1)$ and $\varphi(0.2)$

$h = \frac{x^* - x_0}{m} = 0.1. \Rightarrow \frac{x^* - x_0}{m} = \frac{1}{10} \Rightarrow \frac{1 - x_0}{n} = \frac{1}{10}$

$y' = 1 + xy^2 = g(x, y)$

$y(0) = 0 \Rightarrow y'(0) = 0 \Rightarrow$

$u = g(x_0, y_0)$

$\begin{cases} x_{k+1} = x_k + 0.1 \\ y_{k+1} = y_k + 0.1 \cdot (1 + x_k y_k^2) \end{cases}$

$y_{k+1} = y_k + 0.1 + 0.1 \cdot x_k + 0.1 \cdot y_k^2$

→ Numerical formula: $h = \frac{x^* - x_0}{m}$

$\begin{cases} x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}$ $, k = \overline{0, m-1}$

a) $h > 0$ given

$\begin{cases} x_0 = 0 \\ y_0 = 0 \end{cases}$

Fix x^* $\Rightarrow m = \frac{x^*}{h}$, $g(x, y) = 1 + xy^2$

$g : \mathbb{R}^2 \to \mathbb{R}$ is a C^1 function. $, k = \overline{0, m-1}$

b) $x^* = 1$

b) $x^* = 1 \Rightarrow m = \frac{x^*}{h} = \frac{1}{0.1} = \frac{1}{\frac{1}{10}} = 10.$

$$\begin{cases} x_{k+1} = x_k + h \\ x_0 = 0 \end{cases} \Rightarrow x_{k+1} = x_k + 0.1.$$

$$\Rightarrow x_k = k \cdot h = \frac{k}{10}$$

$x_1 = 0.1.$

$x_2 = 0.2.$

$y_1 \approx \varphi(0.1)$

$y_2 \approx \varphi(0.2)$

$y_1 = y_0 + \frac{1}{10}(1 + x_0 y_0^2) = 0 + 0.1 \cdot (1+0) = 0.1.$

$y_2 = y_1 + 0.1 \cdot (1 + x_1 y_1^2) = 0.1 + 0.1 \cdot (1 + 0.1 \cdot 0.1^2) =$

$= 0.2 + (0.1)^3 = 0.1 + 0.1001 = 0.2001$

Ex2: Consider the IVP $\begin{cases} y' = y \\ y(0) = 1 \end{cases}$. We know that the unique sol: $\varphi: \mathbb{R} \to \mathbb{R}$
$\varphi(x) = e^x$

a) Apply the Euler's numerical formula for this IVP with a ct. step size $h > 0$ and on the interval $[0, x^*]$, where x^* fixed.

b) Prove that $y_k = (1+h)^k$;

c) Fix x^*, let $m = \frac{x^*}{h}$. Prove that $\lim_{m \to \infty} y_m = e^{x^*}$

Sol.

a) $$\begin{cases} h = \frac{x^* - x_0}{m} \\ x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}, k = \overline{0, m-1}$$

~~$x_0 = 0, y_0 = 0$~~

x^* fixed $\Rightarrow m = \frac{x^* - x_0}{h}$

$g(x,y) = y$

$$\Rightarrow \begin{cases} x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot y_k = y_k + h \cdot y_k \Rightarrow \underline{(h+1) y_k = y_{k+1}} \end{cases}$$

$y_0 = 1, y_2 = 1, \dots, y_k = 1, x_0 = 0$

$(y_1 = y_0 = 1 = \dots = y_k)$

)
$g(x,y)=y$

$\left.\begin{array}{l}\boxed{y(x_0)=y_0} \\ y(0)=1\end{array}\right\} \Rightarrow \begin{array}{l} x_0=0 \\ y_0=1\end{array}$

x^* fixed

$h>0$ fixed

$$m=\frac{x^*-x_0}{h}=\frac{x^*-0}{h}=\frac{x^*}{h}$$

$$\begin{cases} x_{k+1}=x_k+h \\ y_{k+1}=y_k+h\cdot y_k \end{cases}$$

b) $$\begin{cases} x_{k+1}=x_k+h, & k=\overline{0,m-1}, \quad x_0=0 \\ y_{k+1}=(1+h)y_k, & y_0=1 \end{cases}$$

? $x_k=k\cdot h$

$y_k=(1+h)^k$ (induction)

c)

$[0, x^*]$

$$y_m \approx e(x_m)=e(x^*)=e^{x^*}$$

$$y_m=\Big(1+\underbrace{\frac{x^*}{m}}_{h}\Big)^m$$

$$\lim_m y_m=\lim_m\left(1+\frac{x^*}{m}\right)^m=e^{x^*}$$

Ex3. Find a range of values for h s.t. the attractor equilibrium point of $\dot{x}=x^2+5x+6$ is also an attractor fixed point of the discrete dym system associated with the Euler's numerical formula with step size $h>0$ for the given d.e.

$\dot{x}=x^2+5x+6=0$

$\Delta=25-4\cdot 6=1 \Rightarrow x_{1,2}=\frac{-5\pm 1}{2}$ $\nearrow x_1^*=-2<0$, $\searrow x_2^*=-3<0$

(First-order scalar non-linear d.e.)

→ Step1: find the attractor eq. point (if ∃)

Theorem: The linearization method for

Let $f\in C^1(\mathbb{R})$ and $\eta^*\in\mathbb{R}$ s.t. $f(\eta^*)=0$ if $\nearrow f'(\eta^*)<0 \Rightarrow$ attr. $\searrow f'(\eta^*)>0 \Rightarrow$ rep.

$g'=2x+5$

$g'(-2)=1$

$g'(-3)=-1<0 \Rightarrow -3$ attr.

→ Step 2: numerical Euler formula.

$\left.\begin{array}{l} x(t) \\ y(x) \end{array}\right\}$ ≠ notations, rewrite:

~~The Euler's numerical formula for $x' = g(t,x)$, $x(t_0) = x_0$.~~

$x = x(t)$ in $\dot{x} = x^2 + 5x + 6$ $\xrightarrow{\text{we change to}}$ $y = y(x)$ and $y' = y^2 + 5y + 6$.

$g(\cancel{x}, y) = y^2 + 5y + 6$

$$\begin{cases} x_{k+1} = x_k + h, & x_0 \\ y_{k+1} = y_k + h \cdot g(x_0, y_0) = \underline{y_k + h \cdot (y_k^2 + 5y_k + 6)}, & y_0 \end{cases}$$

$h > 0$

$x_0, y_0 \in \mathbb{R}$ fixed

$\Rightarrow y_{k+1} = f(y_k)$, y_0 fixed.

$f(y) = y + h \cdot g(y)$

$f(-3) = -3 + h \cdot \underbrace{g(-3)}_{0} = -3 \Rightarrow -3$ fixed point. in f.

$y_{k+1} = \lambda \cdot y_k$, y_0 given

$\lambda \neq 0$

$y_1 = \lambda \cdot y_0$

$y_2 = \lambda \cdot y_1 = \lambda^2 \cdot y_0$

$\vdots$

$y_{k+1} = \lambda^{k+1} \cdot y_0$

$\lim_{k \to \infty} y_k = 0$ for $\lambda \in (-1, 1)$ given range for λ

$2^k \to \infty$

$\left(\frac{1}{2}\right)^k \to 0$

fixed point

Theorem: Let $f \in C^1(\mathbb{R})$ and $\eta^* \in \mathbb{R}$ s.t. $f(\eta^*) = \eta^*$.
If $|f(\eta^*)| < 1$ then the fixed point η^* of $y_{k+1} = f(y_k)$ is an attractor.

fixed point

$f'(y) = 1 + h \cdot g'(y) \to f'(-3) = 1 + h \cdot \underbrace{g'(-3)}_{-1} =$

$g(y) = y^2 + 5y + 6$

$g'(y) = 2y + 5$

$g'(-3) = -1$

$f'(-3) = 1 + h \cdot g'(-3) = 1 - h$

if $|f'(-3)| < 1$ then -3 is an attractor for $y_{k+1} = f(y_k)$

$|1-h| < 1$

$\Leftrightarrow -1 < 1-h < 1 \quad |-1$

$\Rightarrow -2 < -h < 0$

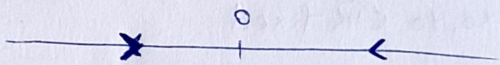
$\Rightarrow h \in (0,2)$

Ex: $\dot{x} = -x^3$

a) Reading the phase portrait, note that the equil. point $\eta^* = 0$ is an attractor

$\dot{x} = 0 \Rightarrow -x^3 = 0 \Leftrightarrow x^* = 0.$

$f' = -3x^2$, $f'' = -6x$?



$f'(0) = 0$ $\quad f'(-1) \leq 0$, $f'(1) < 0$ $\quad \Rightarrow 0$ attractor. ?

b) Let $f(x) = -x^3$. Note that $f(0) = 0$ and $f'(0) = 0$

Ex: $x_{k+1} = x_k - x_k^3$

~~Riemann~~ Remark: Let $x_{k+1} = f(x_k)$ with $f \in C^1(\mathbb{R})$. Let η^* be s.t. $f(\eta^*) = \eta^*$. There exists a function f with a fixed point η^* s.t. η^* attractor and $|f'(\eta^*)| = 1$

! No attr. for discrete systems for exam!